

All Along the dwatch Tower

DTrace tool for the masses

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FreeBSD

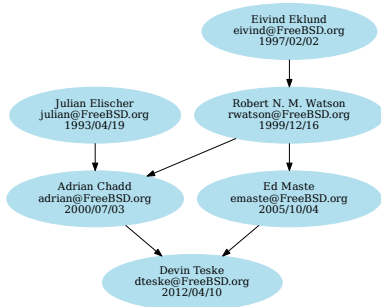
BSDCan Conference

University of Ottawa, Ontario

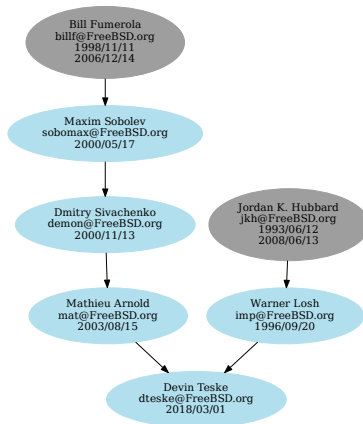
June 8-9, 2018

Who Am I?

src committer



ports committer



Some FreeBSD Contributions (1 of 6)

Boot Menu

FreeBSD 6



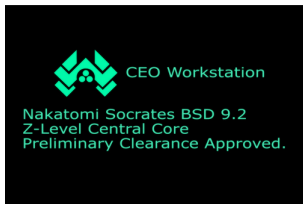
FreeBSD 11



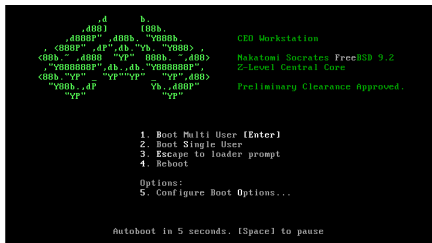
Some FreeBSD Contributions (2 of 6)

Easter Egg

Die Hard (1988)



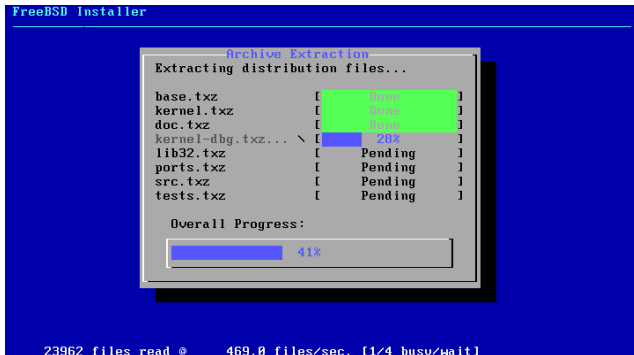
FreeBSD 9.2-RC1 (2013)



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Some FreeBSD Contributions (3 of 6)

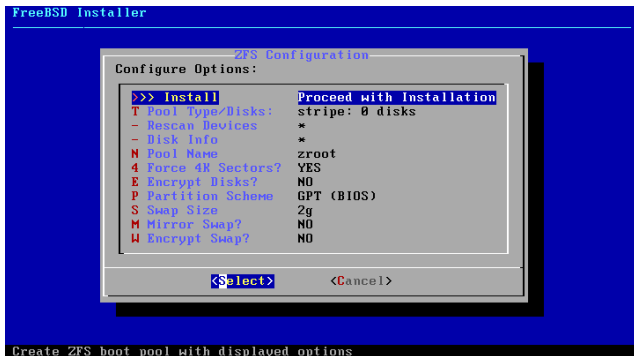
bsdinstall distextract



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Some FreeBSD Contributions (4 of 6)

bsdinstall **zfsboot** (with **allanjude@**)



Some FreeBSD Contributions (5 of 6)

bsdconfig

```
bsdconfig

Main Menu
If you've already installed FreeBSD, you may use
this menu to customize it somewhat to suit your
particular configuration. Most importantly, you
can use the Packages utility to load extra '3rd
party' software not provided in the base
distributions.

  X Exit
  1 Usage
  2 Documentation installation
  3 Packages
  4 Root Password
  5 Disk Management
  6 Login/Group Management
  7 Console
  8 Timezone

  (+) 56%

< OK > <Exit bsdconfig> < Help >

Exit bsdconfig
```

Some FreeBSD Contributions (6 of 6)

sysrc

SYSRC(8)

FreeBSD System Manager's Manual

SYSRC(8)

NAME

`sysrc` - safely edit system rc files

SYNOPSIS

```
sysrc [-cdDeFhinNqvX] [-s name] [-f file] [-j jail | -R dir]
      name[+[|-]=value] ...
sysrc [-cdDeFhinNqvX] [-s name] [-f file] [-j jail | -R dir] -a | -A
sysrc [-E] [-s name] [-f file] -l
sysrc [-eEqv] -L [name ...]
```

DESCRIPTION

The `sysrc` utility retrieves `rc.conf(5)` variables from the collection of system rc files and allows processes with appropriate privilege to change values in a safe and effective manner.

The following options are available:

```
-a          Dump a list of all non-default configuration variables.
-A          Dump a list of all configuration variables (incl. defaults).
:           :
:           :
```



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Today's Talk

3 topics to cover today

- ▶ **DTrace**
- ▶ My new tool - **dwatch**
- ▶ **New ports** (Live Demo)
 - ▶ sysutils/dwatch-pwsnoop
 - ▶ sysutils/dwatch-gource



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DTrace



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On the Topic of DTrace

Overview

- ▶ The Mascot
- ▶ What Can DTrace Do?
- ▶ Brief History and About

Using DTrace (in-depth) - 3 Parts

- ▶ Part 1 - **Probes**
- ▶ Part 2 - **Actions**
- ▶ Part 3 - **Predicates**



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DTrace Mascot

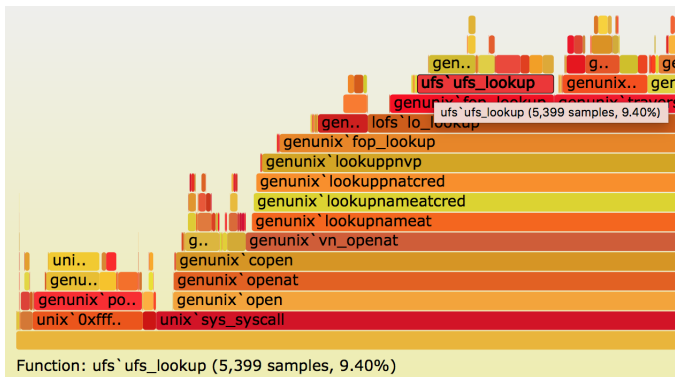
DTrace **ponycorn** (both **pony** *and* **unicorn**!)



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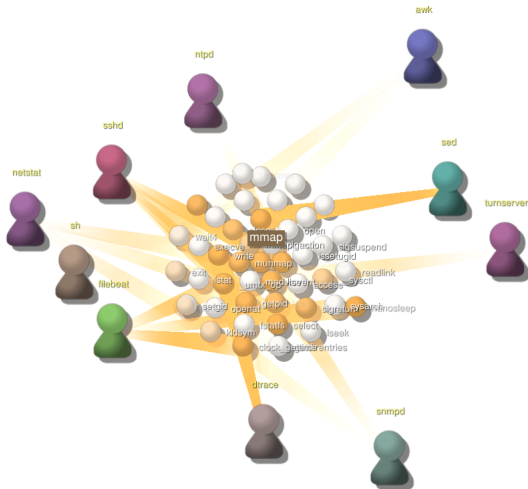
What Can DTrace Do? (1 of 4)

Provide data for interactive **flame graphs** (pictured)



What Can DTrace Do? (2 of 4)

Feed data visualizers such as **gource** (pictured)



What Can DTrace Do? (3 of 4)

Trap arguments from **syscalls**, **functions**, and so much more



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What Can DTrace Do? (4 of 4)

Snoop on all the things (pictured)



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Brief [15-year] History of DTrace Support

- ▶ Nov 2003 – Solaris 10 [**DTrace is born**]
- ▶ Oct 2007 – Ported to Mac OS X 10.5
- ▶ Apr 2008 – dtrace4linux is born
- ▶ Jan 2009 – Ported to **FreeBSD 7.1**
- ▶ Feb 2010 – Ported to NetBSD
- ▶ Dec 2012 – Ported to Oracle Linux 6
- ▶ Feb 2018 – Oracle relicenses DTrace under GPL



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About DTrace

- ▶ Own “D” language
- ▶ Micro virtual machine
- ▶ Conservative probe point selection

– Adam Leventhal [[@ahl](#)], DTrace Co-Creator
[<http://dtrace.org/blogs/ahl/page/11/>]



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Part 1: Probes



Part 1: Basic DTrace Syntax

(as root) **dtrace -n <probe>**

trace a probe

(as root) **dtrace -n <probe>, <probe>, ...**

trace multiple probes



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Part 1: What Is a DTrace Probe?

<probe> is a **colon-separated string**

representing *kernel* or *userland* points to trace

$\langle probe \rangle = 4\text{-Tuple} = \text{provider} : \text{module} : \text{function} : \text{name}$

Example: `vfs:vop:vop_create:entry`



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Part 1: DTrace Output (1 of 5)

(as root) **dtrace -n** **vfs:vop:vop_create:entry**

dtrace: description 'vfs:vop:vop_create:entry' matched 1 probe

CPU	ID	FUNCTION:NAME
3	61206	vop_create:entry
4	61206	vop_create:entry
0	61206	vop_create:entry
3	61206	vop_create:entry
2	61206	vop_create:entry
0	61206	vop_create:entry
2	61206	vop_create:entry
2	61206	vop_create:entry
7	61206	vop_create:entry
:	:	:

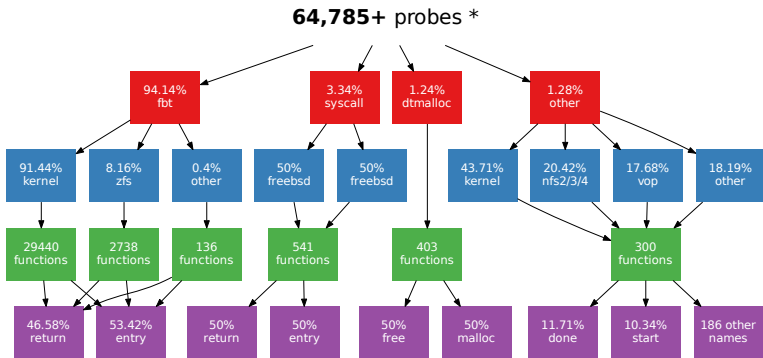


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Part 1: DTrace Probes

How many probes are there?

<probe> =  or simply :



* On FreeBSD 11.0-RELEASE-p1 with dtraceall.ko loaded

Part 1: DTrace Probe Wildcards

`provider`, `module`, `function`, or `name` can use:

- ▶ *Blank* for wildcard
`vfs::vop_create:entry`
- ▶ `*` for wildcard (requires quoting)
`vfs:*.vop_create:entry`
- ▶ `*` for patterns (no regex):
 - ▶ `vfs:vop:*attr:entry`
 - ▶ `syscall::read*:entry`
 - ▶ `vfs::*write*:entry`



Part 1: DTrace Output (2 of 5)

(as root) **dtrace -n "vfs::*write*:entry"**

dtrace: description 'vfs::*write*:entry' matched 4 probes

CPU	ID	FUNCTION:NAME
2	61268	vop_getwritemount:entry
2	61228	vop_write:entry
4	61328	vop_get_writecount:entry
4	61328	vop_get_writecount:entry
4	61268	vop_getwritemount:entry
4	61268	vop_getwritemount:entry
4	61268	vop_getwritemount:entry
4	61268	vop_getwritemount:entry
4	61268	vop_getwritemount:entry
:	:	:



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Part 1: DTrace Multi-Probe Examples

Selecting multiple **different** probes ...

read(2) and **write(2)** syscalls

- ▶ `syscall::read:entry, syscall::write:entry`
2 probes

vfs and **syscall** providers

- ▶ `vfs:::, syscall:::`
2324 probes



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Part 1: DTrace Output (3 of 5)

(as root) **dtrace -n vfs:::,syscall:::**

dtrace: description 'vfs:::,syscall:::' matched 2324 probes

CPU	ID	FUNCTION:NAME
6	62594	ioctl:return
4	60917	lookup:entry
4	61260	vop_lock1:entry
4	61261	vop_lock1:return
6	62593	ioctl:entry
4	61200	vop_islocked:entry
4	61201	vop_islocked:return
4	61202	vop_lookup:entry
4	61216	vop_access:entry
:	:	:



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Part 1: Intermediate DTrace Syntax

(as root) **dtrace** <probe_type> <probe>[,...]

more than just -n and 4-tuples

*tracing an entire **provider**, **module**, **function**, or **name***



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Part 1: How is DTrace Probe Type Used?

<probe_type>

- ▶ Includes -n
- ▶ Must always precede <probe>
- ▶ Allows shorter <probe>'s (3-tuple and less)
- ▶ Applies to all probes



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Part 1: DTrace Probe Type

-n name or 4-tuple (prov:mod:func:name)

▶ -n tick $\Leftrightarrow$ -n :::tick

▶ -n proc:::create,on-cpu $\Leftrightarrow$ -n proc:::create,:::oncpu

-f function

▶ -f log $\Leftrightarrow$ -n ::log:

▶ -f vfs::zap $\Leftrightarrow$ -n vfs::zap:

-m module

▶ -m zfs $\Leftrightarrow$ -n :zfs::

▶ -m fbt:zfs $\Leftrightarrow$ -n fbt:zfs::

-P provider

▶ -P syscall $\Leftrightarrow$ -n syscall:::



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Part 1: DTrace Output (4 of 5)

(as root) **dtrace -m zfs**

CPU	ID	FUNCTION:NAME
1	55893	arc_available_memory:entry
1	61357	none:arc-available_memory
1	55894	arc_available_memory:return
1	55889	arc_adjust:entry
1	55891	arc_adjust_type:entry
1	58175	multilist_get_random_index:entry
1	58352	spa_get_random:entry
1	58353	spa_get_random:return
1	58175	multilist_get_random_index:entry
:	:	:



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Part 1: Listing DTrace Probes

List all probes ...

- ▶ (as root) **dtrace -ln :**

List matching probes ...

- ▶ (as root) **dtrace -lP syscall**
- ▶ (as root) **dtrace -ln "::exec*:entry"**



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Part 1: DTrace Output (5 of 5)

(as root) **dtrace -lP syscall**

ID	PROVIDER	MODULE	FUNCTION	NAME
61403	syscall	freebsd32	syscall	entry
61404	syscall	freebsd32	syscall	return
61405	syscall	freebsd32	exit	entry
61406	syscall	freebsd32	exit	return
61407	syscall	freebsd32	fork	entry
:	:	:	:	:
63562	syscall	freebsd	utimensat	return
63563	syscall	freebsd	numa_getaffinity	entry
63564	syscall	freebsd	numa_getaffinity	return
63565	syscall	freebsd	numa_setaffinity	entry
63566	syscall	freebsd	numa_setaffinity	return



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Part 1: DTrace Probe ID

Probe ID can be used to list probe

(as root) **dtrace -li 61200**

ID	PROVIDER	MODULE	FUNCTION	NAME
<u>61200</u>	vfs	vop	vop_islocked	entry



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Part 2: Actions



Part 2: Advanced DTrace Syntax

(as root) **dtrace -n "<probe> { actions }"**

*more than just cpu, id, **probfunc**, and **probename***



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Part 2: DTrace Default Action

Default action is to print **4 globals**

- ▶ `cpu`
- ▶ `id` [of the probe]
- ▶ `probfunc`
- ▶ `probename`

Default action with **-F** is to print **2 globals**

- ▶ `cpu`
- ▶ `probfunc`



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Part 2: Other DTrace Globals

Globals **NOT** displayed by DTrace (unless asked for)

probeprov = DTrace probe provider

probemod = DTrace probe module

execname = executable name of current process

timestamp = UNIX epoch in UTC with μs precision

walltimestamp = Traditional UNIX epoch

pid = current process id

uid = current user id

gid = current group id

jid = jail id of current process (recently added)

tid = current thread id

curthread = struct thread from /usr/include/sys/proc.h



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Part 2: DTrace Output (1 of 3)

(as root) `dtrace -n 'syscall::: { printf("%s", execname) }'`

dtrace: description 'syscall::: ' matched 2164 probes

CPU	ID	FUNCTION:NAME
4	62594	ioctl:return dtrace
7	63194	kevent:return turnserver
4	62593	ioctl:entry dtrace
4	62594	ioctl:return dtrace
4	62593	ioctl:entry dtrace
4	62594	ioctl:return dtrace
7	63193	kevent:entry turnserver
4	63147	sigprocmask:entry dtrace
4	63148	sigprocmask:return dtrace
:	:	:



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Part 2: DTrace Output (2 of 3)

```
(as root) dtrace -Fm 'zfs {  
    printf("%s", curthread->td_name)  
}'
```

dtrace: description 'zfs' matched 5024 probes

CPU FUNCTION

4	-> arc_do_user_evicts	arc_user_evicts_thr
4	-> arc_kstat_update	arc_user_evicts_thr
4	<- arc_kstat_update	arc_user_evicts_thr
6	-> zio_root	trim zroot
6	-> zio_create	trim zroot
6	-> list_create	trim zroot
6	<- list_create	trim zroot
6	-> list_create	trim zroot
6	<- list_create	trim zroot
:	:	:



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Part 2: Probe Arguments (1 of 2)

Combine -l with -v to list probe arguments

(as root) `dtrace -lvf vop_islocked | awk 'NF==5;/args/,!NF'`

ID	PROVIDER	MODULE	FUNCTION	NAME
61200	vfs	vop	vop_islocked	entry
			args[0]:	struct vnode *
			args[1]:	struct vop_islocked_args *
61201	vfs	vop	vop_islocked	return
			args[0]:	struct vnode *
			args[1]:	struct vop_islocked_args *
			args[2]:	int



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Part 2: Probe Arguments (2 of 2)

Cross-reference with man-pages to build actions

(as root) **dtrace -lvn freebsd:kill:entry | awk 'NF==5;/args/,!NF'**

ID	PROVIDER	MODULE	FUNCTION	NAME
62559	syscall	freebsd	kill	entry
		args[0]:	<u>int</u>	
		args[1]:	<u>int</u>	

man -P "col -b" 2 kill | grep '');

```
kill(pid_t pid, int sig);
```



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Part 2: DTrace Output (3 of 3)

```
(as root) dtrace -n 'kill:entry {  
    printf("pid %i sig %i", arg0, arg1)  
}'
```

```
dtrace: description '::kill:entry' matched 2 probes
```

CPU	ID	FUNCTION:NAME
4	62559	kill:entry pid 62605 sig 0
5	62559	kill:entry pid 62763 sig 2
3	62559	kill:entry pid 62604 sig 0
6	62559	kill:entry pid -62779 sig 15
6	62559	kill:entry pid -62779 sig 19
:	:	:



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Part 3: Predicates (aka Test Conditions)



Part 3: Expert DTrace Syntax

(as root) **dtrace -n "<probe> / tests / { actions }"**

actions only when you want them



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Part 3: DTrace Output (1 of 1)

```
(as root) dtrace -n 'proc:::exec /execname == "bash"/ {  
    printf("fork %u", pid)  
}'
```

```
dtrace: description 'proc:::exec ' matched 1 probe  
CPU      ID      FUNCTION:NAME  
  0  60824      none:exec fork 62990  
  0  60824      none:exec fork 62991  
  7  60824      none:exec fork 62992  
  1  60824      none:exec fork 62993  
  1  60824      none:exec fork 63004  
  0  60824      none:exec fork 63009  
  1  60824      none:exec fork 63026  
  1  60824      none:exec fork 63027  
  5  60824      none:exec fork 63038  
  :      :      :      :      :      :
```



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DTrace Review

dtrace <probe_type> "<probe> [/tests/] { [actions] }" ...

- ▶ Probe, Tests, & Actions must be combined into single arg
- ▶ Default action gives little information
- ▶ One-liners frustrating for people that cannot edit "D"



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dwatch



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On the Topic of dwatch

Overview

- ▶ The Mascot
- ▶ About
- ▶ What dwatch does different vs well vs poorly

Using dwatch (in-depth) - 3 Parts

- ▶ Part 1 - **Probes**
- ▶ Part 2 - **Actions and Predicates**
- ▶ Part 3 - **Profiles**



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dwatch Mascot

“Peaches” the dwatch **ponycorn**



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Available in FreeBSD base (/usr/sbin)

- ▶ 11.2-RELEASE (11.2-RC1 or later)
- ▶ 12.0-CURRENT (Jun 1, 2018 or later)



About dwatch

- ▶ Powerful DTrace wrapper
- ▶ **Intelligent** probe point selection
- ▶ New globals
- ▶ Default action displays different info
- ▶ Profiles
- ▶ sudo(8) integration



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What dwatch Does Different

Parameterization

of Probe, Predicate, and Action

DTrace

makes you pass all of them as a *single argument*

dwatch

separates them for improved usage



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What dwatch Does Well

Filtering and *regex*

- ▶ process id → -p pid|regex
- ▶ user → -u uid|name|regex
- ▶ group → -g gid|name|regex
- ▶ jail → -j jid|name
- ▶ execname → -k [*]name[*] **or** -z regex
- ▶ anything → -r regex

Data Sampling (for example, via cron)

- ▶ by count → -N num
- ▶ by timeout → -T duration



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What dwatch Does **Not** Do Well

DTrace Aggregations require debug passthrough (see below)

```
dwatch _dE '@[probfunc]=count()' syscall | dtrace -s /dev/stdin
```

DTrace needs to be foreground to catch Ctrl-C

and

Some features rely on dwatch being foreground
and thus **may not work** in debug passthrough



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Part 1: Probes



Part 1: Basic dwatch Syntax

dwatch <probe>

watch a probe

dwatch <probe>,<probe>,...

watch multiple probes



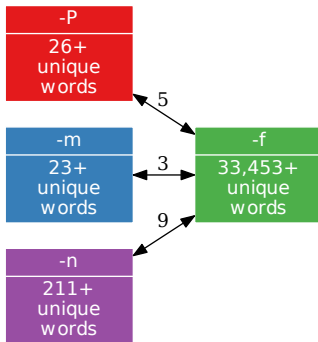
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Part 1: DTrace Probe Type Overlap (1 of 4)

dwatch makes **<probe_type>** optional

33,721+ (99.95%) of single words are unique

only 17 words (0.05%) overlap between categories



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Part 1: DTrace Probe Type Overlap (2 of 4)

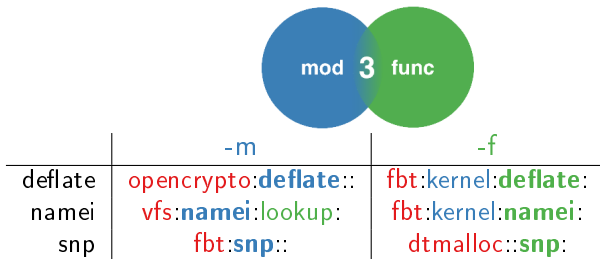


	-P	-f
fbt	fbt :::	dtmalloc:: fbt :
proc	proc ::kernel::	dtmalloc:: proc :
racct	racct ::kernel::	racct::kernel: racct :
random	random :::	fbt::kernel: random :
syscall	syscall :::	syscall:: syscall :



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Part 1: DTrace Probe Type Overlap (3 of 4)



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Part 1: DTrace Probe Type Overlap (4 of 4)



	-f	-n
create	<code>nfsc1::create:</code>	<code>:kernel::create</code>
exit	<code>syscall::exit:</code>	<code>proc:kernel:none:exit</code>
flush	<code>nfsc1::flush:done</code>	<code>xbb:kernel::flush</code>
free	<code>fbt:kernel::free:</code>	<code>dtmalloc::free</code>
init	<code>proc:kernel::init:</code>	<code>sctp:cwnd:net:init</code>
malloc	<code>fbt:kernel::malloc:</code>	<code>dtmalloc::malloc</code>
read	<code>::read:</code>	<code>xbb:kernel::read</code>
wakeup	<code>fbt:kernel::wakeup:</code>	<code>sched:kernel:none:wakeup</code>
write	<code>::write:</code>	<code>xbb:kernel::write</code>



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Part 1: dwatch Default Action (1 of 3)

Default action is to print **6 globals**

- ▶ walltimestamp in "%Y %b %e %T" format
- ▶ uid
- ▶ gid
- ▶ execname
- ▶ pid
- ▶ curthread cmdline arguments



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Part 1: dwatch Output (1 of 3)

dwatch vop_create

```
INFO Watching 'vfs:vop:vop_create:entry' ...
2018 Jun  2 23:45:17 604.604 touch[57667]: touch foo
2018 Jun  2 23:45:23 604.604 touch[57685]: touch bar
2018 Jun  2 23:45:43 0.0 ruby23[57697]: ruby23: chef-client worker: ppid=886;st
2018 Jun  2 23:45:43 0.0 ruby23[57697]: ruby23: chef-client worker: ppid=886;st
2018 Jun  2 23:45:43 0.0 ruby23[57697]: ruby23: chef-client worker: ppid=886;st
2018 Jun  2 23:45:43 0.0 ruby23[57697]: ruby23: chef-client worker: ppid=886;st
2018 Jun  2 23:45:43 0.0 pkg[57844]: pkg rquery %v nagios-check_zpools
2018 Jun  2 23:45:43 0.0 pkg[57844]: pkg rquery %v nagios-check_zpools
2018 Jun  2 23:45:43 0.0 pkg[57844]: pkg rquery %v nagios-check_zpools
2018 Jun  2 23:45:43 0.0 pkg[57844]: pkg rquery %v nagios-check_zpools
      :           :           :           :
```

Part 1: dwatch Default Action (2 of 3)

Default action with **-F** is to print **10 globals**

Previously mentioned 6 plus ...

- ▶ `probeprov`
- ▶ `probemod`
- ▶ `probefunc`
- ▶ `probename`



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Part 1: dwatch Output (2 of 3)

dwatch -F proc

INFO Watching 'proc:::' ...

```
2018 Jun 3 02:54:04 0.0 intr[12]: | proc:kernel:none:signal-send intr
2018 Jun 3 02:54:04 604.604 bash[66077]: -> proc:kernel:ctor:entry -bash
2018 Jun 3 02:54:04 604.604 bash[66077]: <- proc:kernel:ctor:return -bash
2018 Jun 3 02:54:04 604.604 bash[66077]: | proc:kernel:none:create -bash
2018 Jun 3 02:54:04 604.604 bash[66160]: | proc:kernel:none:exit -bash
2018 Jun 3 02:54:04 604.604 bash[66160]: | proc:kernel:none:signal-send -bash
2018 Jun 3 02:54:04 604.604 bash[66077]: -> proc:kernel:dtor:entry -bash
2018 Jun 3 02:54:04 604.604 bash[66077]: <- proc:kernel:dtor:return -bash
2018 Jun 3 02:54:04 604.604 bash[66077]: -> proc:kernel:ctor:entry -bash
2018 Jun 3 02:54:04 604.604 bash[66077]: <- proc:kernel:ctor:return -bash
      :                :                :                :
```



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Part 1: dwatch Default Action (3 of 3)

-**R** enables a mini process tree!



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Part 1: dwatch Output (2 of 3)

dwatch -R execve

```
INFO Watching 'syscall:freebsd:execve:entry' ...
2018 Jun  4 00:12:22 604.604 sshd[18434]: -bash
    += 01048 0.0 /usr/sbin/sshd
    \ += 18432 0.604 sshd: dteske [priv]
        \ += 18434 604.604 sshd: dteske@pts/1
            \ += 18435 604.604 -bash
2018 Jun  4 00:12:22 604.604 bash[18436]: uname -s
    += 18434 604.604 sshd: dteske@pts/1
    \ += 18435 604.604 -bash
        \ += 18436 604.604 -bash
            \ += 18437 604.604 uname -s
...
...
```



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Part 1: Other DTrace Globals

Globals **NOT** displayed by dwatch (unless asked for)

cpu = active CPU (number)

id = probe ID

tid = current thread id

jid = jail id of current process



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Part 2: Actions and Predicates



Part 2: Intermediate dwatch Syntax

`dwatch -E "actions" <probe>[,...]`

-E can be specified multiple times

`dwatch -t "tests" <probe>[,...]`

-t can be specified multiple times



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Part 2: dwatch Output (1 of 2)

`dwatch -E 'printf("pid %i sig %i", arg0, arg1)' kill`

```
INFO Watching 'syscall::kill:entry' ...
2018 Jun  3 03:29:15 604.604 bash[57550]: pid 57550 sig 0
2018 Jun  3 03:29:23 604.604 bash[68584]: pid 68584 sig 2
2018 Jun  3 03:29:33 604.604 bash[57550]: pid 57549 sig 0
2018 Jun  3 03:29:58 604.604 bash[57550]: pid -68589 sig 15
2018 Jun  3 03:29:58 604.604 bash[57550]: pid -68589 sig 19
      :           :           :           :
```



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Part 2: dwatch Output (2 of 2)

`dwatch -t 'this->caller_execname == "bash"'` **execve**

INFO Watching 'syscall:freebsd:execve:entry' ...

INFO Setting test: this->caller_execname == "bash"

2018 Jun 3 05:26:48 604.604 bash[57550]: ls -AGCF

2018 Jun 3 05:27:02 604.604 bash[57550]: date

2018 Jun 3 05:27:19 604.604 bash[57550]: /bin/pwd

2018 Jun 3 05:27:40 604.604 bash[57550]: id

2018 Jun 3 05:27:45 604.604 bash[57550]: whoami

⋮ ⋮ ⋮ ⋮



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Part 3: Profiles



Part 3: Advanced dwatch Syntax

dwatch -X profile

loads a “canned” solution

dwatch -Q

list available profiles



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Part 3: dwatch Profiles (1 of 8)

dwatch **Profiles** automate:

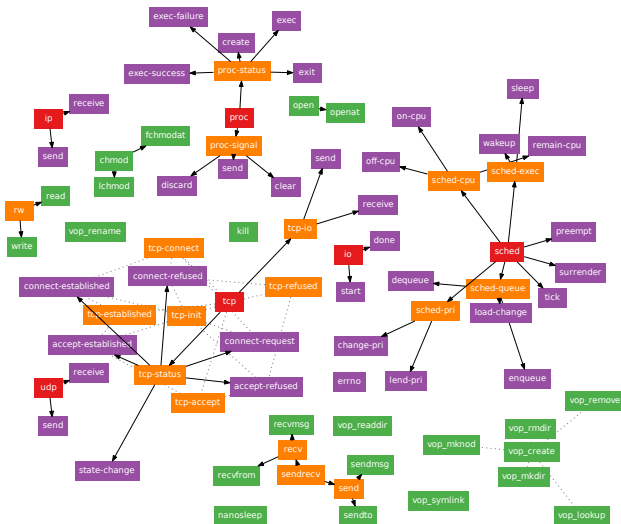
- ▶ Probe selection
- ▶ Data collection
- ▶ Formatting and display



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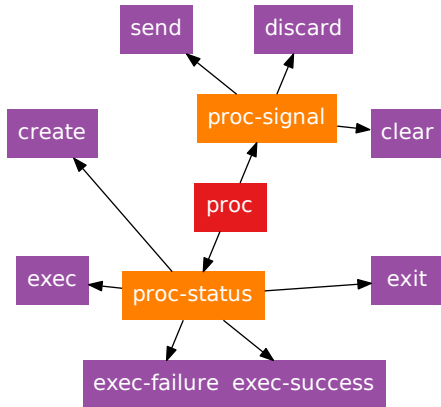
Part 3: dwatch Profiles (2 of 8)

Currently 81 Profiles in base



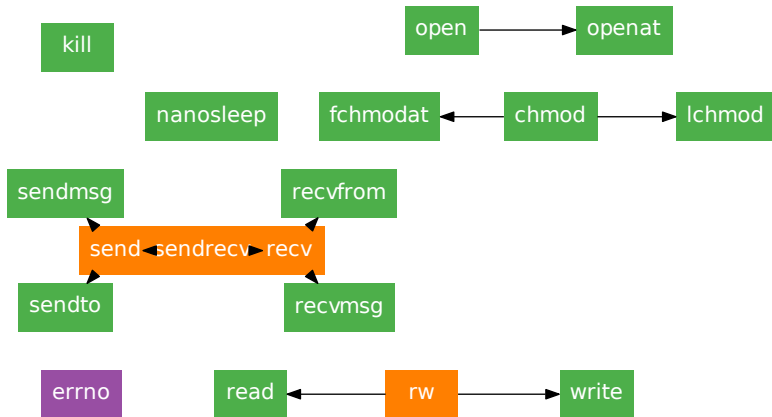
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11 **proc** Profiles for Process Accounting



Part 3: dwatch Profiles (4 of 8)

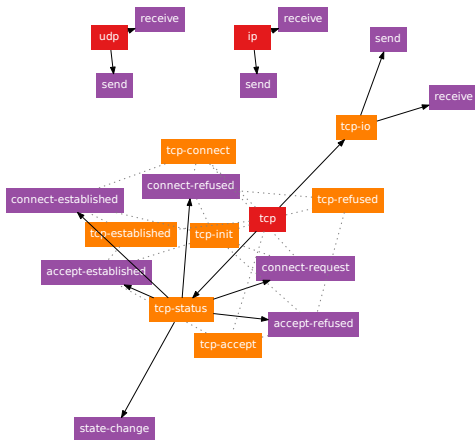
18 **syscall** Profiles



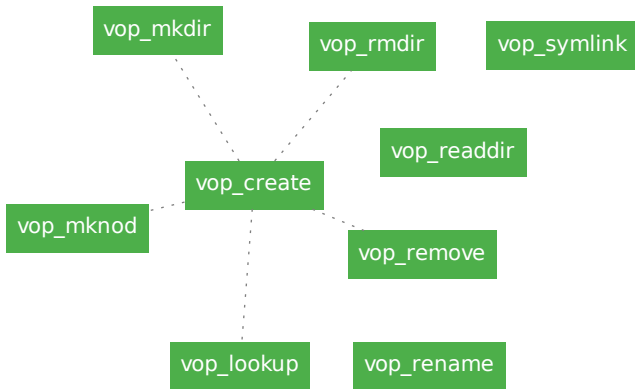
FreeBSD

Part 3: dwatch Profiles (5 of 8)

22 Profiles for Network Monitoring

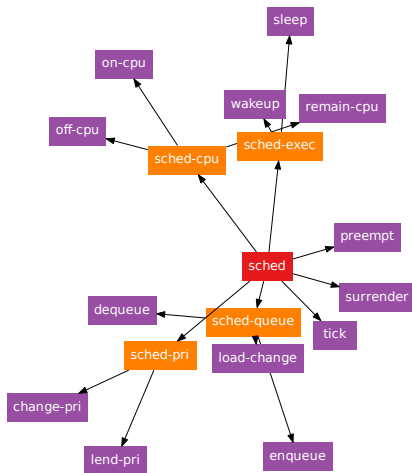


9 **vfs** Profiles for Filesystem Monitoring



Part 3: dwatch Profiles (7 of 8)

18 **sched** Profiles for CPU monitoring



Part 3: dwatch Profiles (8 of 8)

3 **io** Profiles for Disk I/O



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Part 3: dwatch Output (1 of 4)

Use '-r regex' with '-Q' to find profiles

dwatch **-Qr vop**

PROFILES:

vop_create

vop_mknod

vop_rename

vop_lookup

vop_readdir

vop_rmdir

vop_mkdir

vop_remove

vop_symlink



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Part 3: dwatch Output (2 of 4)

`dwatch -X kill`

```
INFO Sourcing kill profile [found in /usr/libexec/dwatch]
INFO Watching 'syscall::kill:entry' ...
2018 Jun 4 00:34:05 604.604 bash[19309]: signal 0 to pid 19309
2018 Jun 4 00:34:21 604.604 bash[19372]: signal 2 to pid 19372
2018 Jun 4 00:34:32 604.604 bash[19309]: signal 0 to pid 19308
2018 Jun 4 00:34:47 604.604 bash[19309]: signal 15 to pid -19725
2018 Jun 4 00:34:47 604.604 bash[19309]: signal 19 to pid -19725
      :           :           :           :
```



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Part 3: dwatch Output (3 of 4)

dwatch -X open -t 'execname != "sendmail"'

INFO Sourcing open profile [found in /usr/libexec/dwatch]

INFO Watching 'syscall::open:entry, syscall::openat:entry' ...

INFO Setting test: execname!="sendmail"

2018 Jun 4 03:11:21 0.0 turnserver[896]: /var/log/turn_2018-06-04.log

2018 Jun 4 03:11:21 0.0 dtrace[26718]: /usr/share/zoneinfo/UTC

2018 Jun 4 03:11:21 0.0 dtrace[26718]: /usr/share/zoneinfo/posixrules

2018 Jun 4 03:11:26 0.0 turnserver[896]: /var/log/turn_2018-06-04.log

2018 Jun 4 03:11:28 604.604 ls[26719]: /etc/libmap.conf

2018 Jun 4 03:11:28 604.604 ls[26719]: /usr/local/etc/libmap.d

2018 Jun 4 03:11:28 604.604 ls[26719]: /usr/local/etc/libmap.d/nginx.c

2018 Jun 4 03:11:28 604.604 ls[26719]: /var/run/ld-elf.so.hints

2018 Jun 4 03:11:28 604.604 ls[26719]: /lib/libxo.so.0

2018 Jun 4 03:11:28 604.604 ls[26719]: /lib/libutil.so.9

⋮ ⋮ ⋮ ⋮



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Part 3: dwatch Output (4 of 4)

`dwatch -X vop_lookup -r /foo/`

```
INFO Sourcing vop_lookup profile [found in /usr/libexec/dwatch]
INFO Watching 'vfs:vop:vop_lookup:entry' ...
INFO Filtering filter: /foo/
2018 Jun  4 03:22:07 604.604 bash[27166]: /usr/home/dteske/foo/.
2018 Jun  4 03:22:07 604.604 bash[19309]: /usr/home/dteske/foo/.
2018 Jun  4 03:22:07 604.604 bash[19309]: /usr/home/dteske/foo/.
2018 Jun  4 03:22:12 604.604 bash[19309]: /usr/home/dteske/foo/.
2018 Jun  4 03:22:12 604.604 touch[27168]: /usr/home/dteske/foo/bar
2018 Jun  4 03:22:12 604.604 touch[27168]: /usr/home/dteske/foo/bar
2018 Jun  4 03:22:31 604.604 mkdir[27169]: /usr/home/dteske/foo/baz
2018 Jun  4 03:22:34 604.604 bash[19309]: /home/dteske/foo/baz
2018 Jun  4 03:22:34 604.604 bash[19309]: /usr/home/dteske/foo/baz
2018 Jun  4 03:22:34 604.604 bash[19309]: /usr/home/dteske/foo/baz
      ⋮                ⋮                ⋮                ⋮
```



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New ports



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New Ports Using dwatch

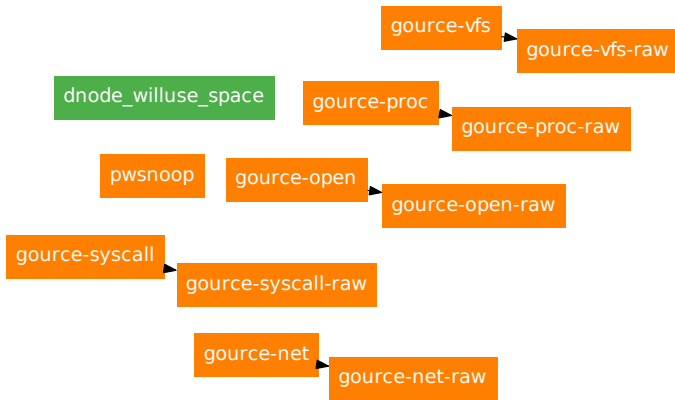
sysutils/**dwatch-pwsnoop**
password snooping

sysutils/**dwatch-gource**
network, process, filesystem, and syscall visualization with *gource*



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More dwatch profiles!



New ports (Live Demo)



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New Ports: Canned Demos

Network Visualization:

- ▶ youtu.be/0l8ZozQKo2w

Syscall Visualization:

- ▶ youtu.be/eYJVJF6ku3c

Process Visualization (single vs. threaded buildworld)

- ▶ youtu.be/2L2QXETIzVI

Filesystem Visualization (chef-client run)

- ▶ youtu.be/spucbh09zT8



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Special Thanks

- ▶ Spouse (hand-drawn “Peaches” mascot)
- ▶ Antti Antinoja (vectorized “Peaches”)
- ▶ FreeBSD reviewers gnn, markj, bdrewery
- ▶ FreeBSD community feedback
- ▶ You! The audience



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Questions?

dteske@FreeBSD.org

| **fraubsd.org/doc/bsdcan/2018/slides**



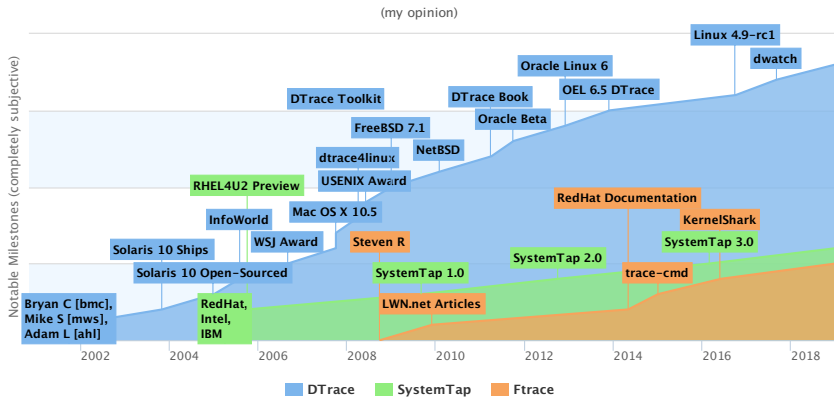
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DTrace and Competitor Timeline



More FreeBSD Fun

Gandi.net sent me rare **T-Shirt** (left)

I thanked them with **ASCII art** (right)

